in shape

e.g. S-orbital is spherical. P orbitals are dumb-bell shaped etc

7

5. State and Explain Pauli's Exclusion principal.

Pauli's Exclusion principal: No two electrons in an atom can have the same set of four identical quantum numbers.

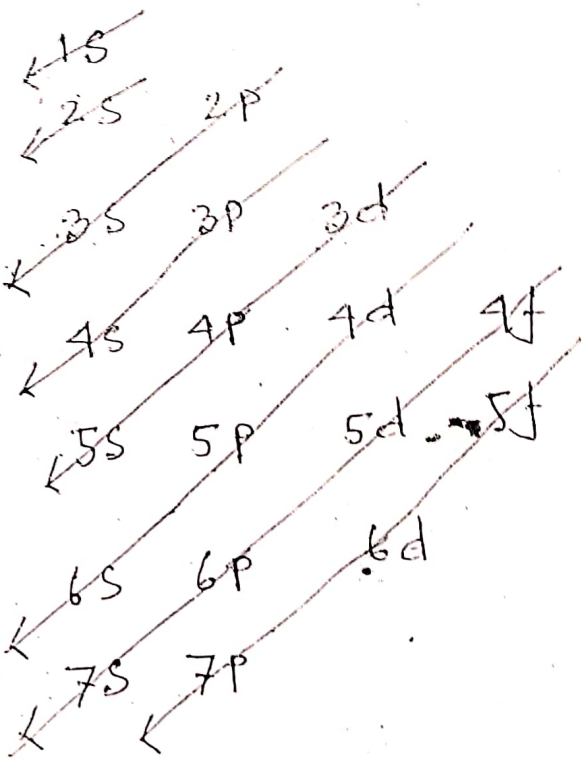
Explanation: Even if two electrons have the same values for n, l and m, they must have different values of s. Thus, every electron in an atom differs from every other electron in total energy and, therefore, there can be as many electrons in a shell as there are possible arrangements of different quantum numbers

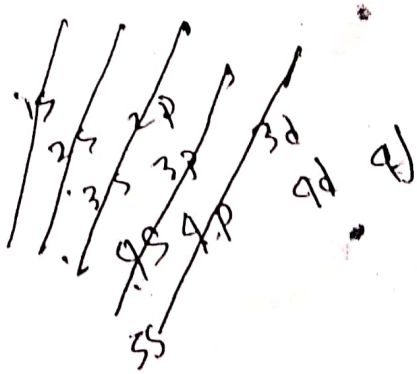
(5)

6. State and explain Aufbau principal.

Aufbau principal: In the ground state of an atom, the electrons tend to occupy the available orbitals in the increasing order of energies, the orbitals of lower energy being filled first.

Explanation: Lower energy orbitals are, therefore, better seats for electrons and better seats are occupied first. Aufbau order of orbitals for feeding in electrons are given below.





The increasing order of energy of various orbitals is as follows.

1S < 2S < 2P < 3S < 3P < 4S < 3d < 4P < 5S < 4d < 5P < 6S .....

7. State and Explain "Hund's" rule.

Hund's rule: Electrons are distributed among the degenerate orbitals of a subshell in such a way as to give the maximum number of unpaired electrons and have the same direction of spin

Explanation:- This rule is based on the fact that electrons, being of the same charge, repel each other and hence try to keep farthere aparet from each other as much as possible. This they do by entering the degenerate orbitals as far as possible. Their entering the orbitals in this manner minimises the inter-electronic repulsion energy of the system and hence we get a stable system.

8. What do you mean by Heisenberg uncertainty principle?

Principle:— It is